Characterizing corn insect protection technology

PIONEER®

ABOVE-GROUND INSECT PROTECTION

		European corn borer#	Southwestern corn borer#	Black cutworm	Corn earworm#	Western bean cutworm*	Fall armyworm#	Herbicide tolerance
Options in the Pioneer® brand portfolio	**Optimum® AcreMax® PRODUCTS**	+++▾	+++▾	++	+	–	++	glyphosate, glufosinate
	Optimum® AcreMax® Leptra® & Optimum® Leptra® PRODUCTS	+++▾	+++▾	+++▾	+++	+++	+++▾	glyphosate, glufosinate
	Optimum® Intrasect® PRODUCTS	+++▾	+++▾	++	+	–	++	glyphosate, glufosinate
	PowerCore® Enlist® corn PRODUCTS	+++▾	+++▾	++	+	–	++▾	2,4-D choline, glyphosate, glufosinate, FOP
	PowerCore® Ultra Enlist® corn PRODUCTS	+++▾	+++▾	+++▾	+++	+++	+++▾	2,4-D choline, glyphosate, glufosinate, FOP
Competitive products	**VT Double PRO® RIB Complete® corn**	+++▾	+++▾	–	+	–	++▾	glyphosate
	Trecepta® corn	+++▾	+++▾	+++	+++	+++	+++▾	glyphosate
	Agrisure® 3120 E-Z Refuge® corn	+++▾	+++▾	++	+	–	++	glyphosate, glufosinate
	Agrisure® 3330 E-Z Refuge® corn	+++▾	+++▾	+++	+++	+++	+++▾	glyphosate, glufosinate

+++ Excellent protection ++ Good protection + Some protection – No activity/not labelled ▾2 or more working modes of action

ABOVE- AND BELOW-GROUND INSECT PROTECTION

		Corn rootworm (western#, northern, Mexican)	European corn borer	Southwestern corn borer#	Black cutworm	Corn earworm#	Western bean cutworm#	Fall armyworm#	Herbicide tolerance
Options in the Pioneer® brand portfolio	Qrome® PRODUCTS	++▼	+++▼	+++▼	++	+	–	++	glyphosate, glufosinate
	Optimum® AcreMax® XTreme PRODUCTS	++▼	+++▼	+++▼	++	+	–	++	glyphosate, glufosinate
	Vorceed® Enlist® corn PRODUCTS	+++▼	+++▼	+++▼	++	+	–	++▼	2,4-D choline, glyphosate, glufosinate, FOP
Competitive products	SmartStax® Technology	++▼	+++▼	+++▼	++	+	–	++▼	glyphosate, glufosinate
	SmartStax® PRO Technology	+++▼	+++▼	+++▼	++	+	–	++▼	glyphosate, glufosinate
	VT4PRO™ with RNAi Technology	++▼	+++▼	+++	+++	+++	+++	+++▼	glyphosate
	Agrisure® Duracade® 5122 E-Z Refuge® corn	++▼	+++▼	+++▼	++	+	–	++	glyphosate, glufosinate

+++ Excellent protection ++ Good protection + Some protection – No activity/not labelled ▼2 or more working modes of action

Characterization from Internal Corteva Agriscience Attributes database.

Efficacy levels based on Corteva Agriscience and/or independent university entomologist results against susceptible insect populations. Product responses can vary by location, pest population, environmental conditions, and agricultural practices.

Please contact your Pioneer sales professional for information and suggestions specific to your operation. Individual results may vary.

Various factors, including pest pressure, reduced susceptibility, and insect resistance in some pest populations may affect efficacy of certain corn technology products in some regions. To help extend durability of these technologies, Corteva Agriscience recommends you implement Integrated Pest Management (IPM) practices such as crop rotation, cultural and biological control tactics (including rotating sources of Bt-protected corn traits), pest scouting, and appropriate use of pest thresholds when employing management practices such as insecticide application.You must also plant the required refuge when using these technologies. Please contact your sales professional or consult with your local university extension for more information regarding insect resistance management guidelines, best management practices and to understand whether there has been a shift in susceptibility or insect resistance with certain pests documented in your area.
* Western bean cutworm has been removed from the Corteva Agriscience product use statement for several corn products that contain Herculex® I (Cry1F) but lack another mode of action for western bean cutworm due to a wide-spread decrease in susceptibility indicating the possibility of field-evolved resistance to Cry1F in most geographies.

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Following burndown, Enlist Duo® and Enlist One® herbicides with Colex-D® technology are the only herbicides containing 2,4-D that are authorized for preemergence and postemergence use with Enlist® crops. Consult Enlist® herbicide labels for weed species controlled. Enlist Duo and Enlist One herbicides are not registered for use or sale in all states and counties; are not registered in AK, CA, CT, HI, ID, MA, ME, MT, NH, NV, OR, RI, UT, VT, WA and WY; and have additional subcounty restrictions in AL, GA, TN and TX, while existing county restrictions still remain in FL. All users must check "Bulletins Live! Two" no earlier than six months before using Enlist One or Enlist Duo. To obtain "Bulletins," consult epa.gov/espp/, call 1-844-447-3813, or email ESPP@epa.gov. You must use the "Bulletin" valid for the month and state and county in which Enlist One or Enlist Duo are being applied. Contact your state pesticide regulatory agency if you have questions about the registration status of Enlist® herbicides in your area. ALWAYS READ AND FOLLOW PESTICIDE LABEL DIRECTIONS. IT IS A VIOLATION OF FEDERAL AND STATE LAW TO USE ANY PESTICIDE PRODUCT OTHER THAN IN ACCORDANCE WITH ITS LABELING. ONLY USE FORMULATIONS THAT ARE SPECIFICALLY LABELED FOR SUCH USE IN THE STATE OF APPLICATION. USE OF PESTICIDE PRODUCTS, INCLUDING, WITHOUT LIMITATION, 2,4-D-CONTAINING PRODUCTS NOT AUTHORIZED FOR USE WITH ENLIST CROPS, MAY RESULT IN OFF-TARGET DAMAGE TO SENSITIVE CROPS/AREAS AND/OR SUSCEPTIBLE PLANTS, IN ADDITION TO CIVIL AND/OR CRIMINAL PENALTIES. Additional product-specific stewardship requirements for Enlist crops, including the Enlist Product Use Guide, can be found at www.traitstewardship.com.

CORTEVA™ agriscience

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